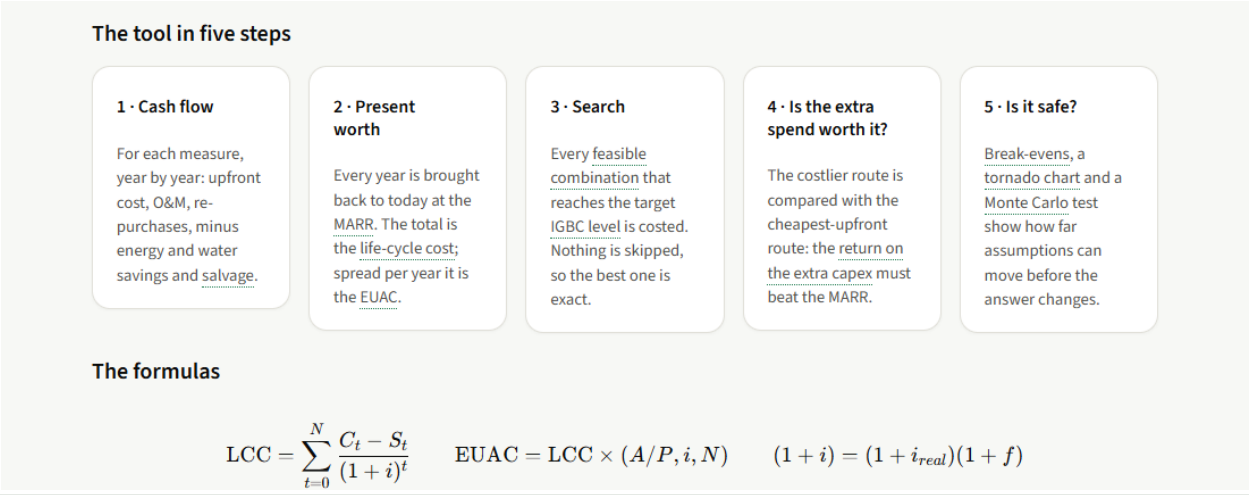
How the tool works

From the input workbook to the recommendation: the method, one worked example, the checks and the limits.

The pipeline in five steps

Step	What happens	Why
1. Cash flow	For each green measure, the tool builds a year-by-year list of money in and out, compared with the ordinary (code-minimum) option.	Cost and savings arrive in different years.
2. Present worth	Every year's amount is brought back to today's rupees at the MARR and added up. That total is the life-cycle cost.	A rupee saved in year 20 is worth less than a rupee spent today.
3. Search	Every legal combination (one option per group, fits the roof, meets mandatory credits, reaches the target IGBC level) is costed.	The exact best answer, not a guess.
4. Is the extra spend worth it?	The costlier route is compared with the cheapest-upfront route; the return on the extra money is compared with the MARR.	Totals alone can mislead; the extra rupees must earn their keep.
5. Is it safe?	Break-even points, a one-at-a-time sensitivity (tornado) and a Monte Carlo simulation test the answer.	Assumptions will not come true exactly.



Step 1: what goes into one measure's cash flow

- **Year 0:** the upfront cost (capex), extra to the ordinary option it replaces.
- **Every year:** maintenance (O&M), rising with general inflation; minus the savings, which are kWh saved × the electricity tariff (rising each year) plus water saved × the water tariff. Equipment slowly loses output (degradation), so savings shrink a little.
- **Solar:** output splits into the share used on site (worth the full tariff) and the share exported (worth the lower export credit).
- **Replacements:** when a measure's life ends inside the study period it is bought again, at an inflated price. A separately wearing part (for example solar inverters) is bought again every part life.
- **Year N (end of study period):** any life left in equipment comes back as salvage value, so options with different lives compare fairly.

Step 2: present worth and cost per year

Life-cycle cost (LCC) = sum over years t = 0..N of ($cost_t$ − $saving_t$) / $(1 + MARR)^t$
Cost per year (EUAC) = LCC × (A/P, MARR, N)
(1 + MARR) = (1 + real rate) × (1 + inflation)

The demo uses MARR 10%, inflation 5% (so the real rate is 4.76%) and N = 25 years. A negative LCC means the measure pays for itself.

A worked example you can check by hand: LED lighting with sensors

Upfront ₹54.0 L, life 12 years (so it is bought again once inside 25 years), no maintenance, saves 108,000 kWh in year 1 at ₹13/kWh = ₹14.04 L, with the tariff rising 5% a year and 1% a year degradation (net growth 3.95%).

Item	Factor form	Present worth
Upfront cost, year 0	P	₹54.0 L
Savings: ₹14.04 L in year 1, growing 3.95% a year	− A_1 × (P/A_1, g = 3.95%, i = 10%, 25) = × 12.511	−₹175.6 L
Re-purchase in year 12 and 24, minus salvage at year 25	sum of cost × (F/P) × (P/F), less salvage × (P/F)	₹33.8 L
Life-cycle cost	sum of the lines	**−₹87.8 L**

The tool's year-by-year engine gives the same figure (−₹87,80,302 from both methods, to the rupee). The tool shows this check live on the Recommendation page, in the "How it works" tab.

Step 3: choosing the combination

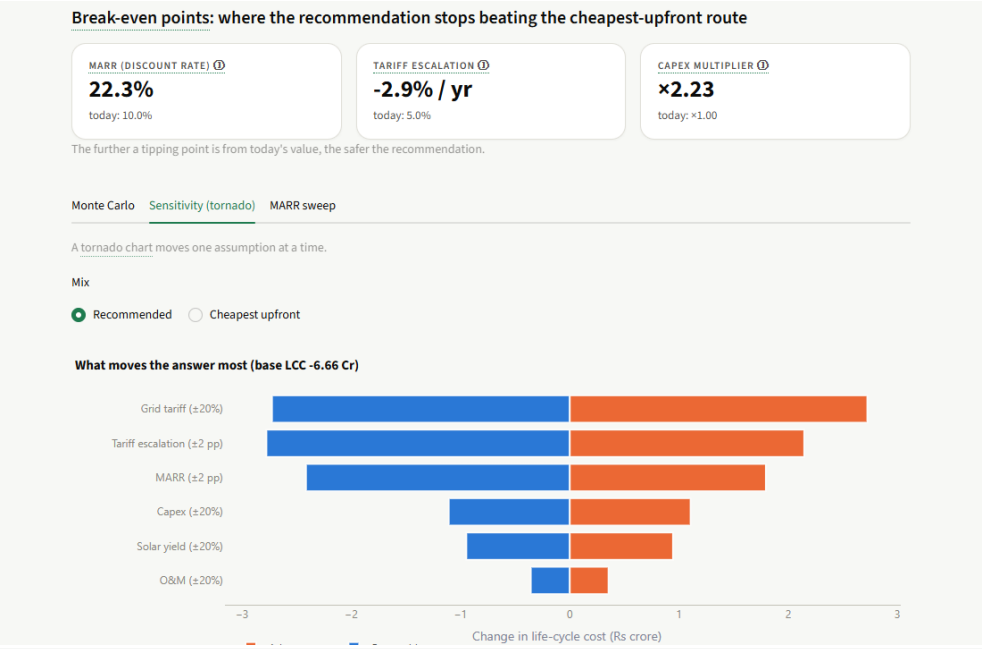
Measures in the same group are alternatives (one HVAC system, one solar size), so a combination is one option per group, or none, plus any independent measures. The demo has 12 measures and 1,152 combinations; all are costed. A combination is **feasible** if it reaches the target IGBC points, meets every mandatory credit and the solar fits the roof. IGBC points are scored from the combined energy and water savings, so nothing is double counted. Two answers are kept: the **cheapest upfront** and the **lowest life-cycle cost**.

Step 4: is the extra spend worth it?

Ranking is always by life-cycle cost, never by rate of return. The rate of return is used for the accept-or-reject question: the extra capex of the recommended route over the cheapest-upfront route has its own cash flow, and the rate it earns (incremental IRR) is compared with the MARR. If that cash flow changes sign more than once, IRR can give several answers, so the tool uses the modified IRR (MIRR) instead. In the demo the extra ₹2.75 Cr earns 13.8% a year against the 10% required, so it is worth it.

Step 5: how safe is it?

Test	What it does	Demo result
Break-even	Finds the MARR, tariff growth and equipment price at which the two routes cost the same.	Flips only above a 22.3% MARR, or if tariffs fall about 2.9% a year, or equipment costs 2.2 times more
Tornado (sensitivity)	Moves one assumption at a time and charts the effect on cost.	Shows which inputs to verify first
MARR sweep	Re-costs both routes across required returns from 4% to 20%.	Lines cross at the break-even MARR
Monte Carlo	3,000 random futures: tariff growth, MARR, equipment price and solar output vary within sensible ranges (fixed seed, so it repeats).	Recommended route wins in 100% of runs



How we checked the tool

- **Hand calculations:** the example above, and the on-screen hand-check, rebuild a measure with textbook interest-factor formulas and match the engine.
- **82 automatic tests:** worked examples from the CE 725 notes (interest factors, present worth, EUAW, IRR, the multiple-IRR case and MIRR), engine hand cases, IGBC scoring, the optimizer, risk, exports and every screen.
- **Textbook slips found:** the tests showed small rounding differences in two worked examples in the course notes, which we recorded.
- **Real browser check:** every page loads in about two seconds with no errors.

Assumptions and limits

- Nominal rupees: prices rise with inflation, and the discount rate is the nominal MARR.
- Base prices are year-1 prices; a replacement uses today's price inflated to the year it is bought.
- Savings of different measures add up, with no interaction between them.
- Owner's pre-tax view; depreciation, tax and financing are outside the model.
- Prices and tariffs are sourced estimates; the IGBC point tables are illustrative.

What is inside the folder

File or folder	Role
app.py, pages/	The screens (Home, Set up, Recommendation, Can we trust it?, Report)
core/finance.py, cashflow.py	The course formulas and the year-by-year engine
core/optimizer.py, igbc.py	The search over combinations and IGBC scoring
core/risk.py, carbon.py	Break-even, tornado, Monte Carlo, CO_2 avoided
core/report.py, exports.py	The memo, PDF report and Excel results
core/hover.py, ui.py	The hover definitions and the interface pieces
data/demo_input.xlsx	The demo inputs, with units and sources
tests/	The 82 automatic tests
launcher.py, Start_GreenLCC.bat	One-click start on your computer